

KHUSHI BHANSALI

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EDUCATION

Cornell University , New York, NY	May 2024
Master of Engineering in Computer Science Merit Scholar	
University of Maryland , College Park, MD	May 2023
Bachelor of Science in Computer Science [Concentration: Machine Learning]	

TECHNICAL SKILLS

Languages: Python, SQL, Java, C#, JavaScript, MATLAB, PySpark | Machine Learning: Transformers, CNN/R-CNN, Autoencoders, Representation Learning, Bayesian Methods, Monte Carlo | Data & Infra: AWS (SageMaker, S3, EC2), Hive, Impala, Linux, Terraform | Creative Tech: Unity, WebXR, Node.js, HTML/CSS | Tools: Git, JIRA, Confluence, Cursor/Claude AI

WORK EXPERIENCE

PNC Bank , Data Scientist	July 2024 - Present
- Owned end-to-end model development, evaluation, backtesting, drift analysis, and production deployment of a Check Deposit Fraud model, preventing \$13M+ in annual losses. - Leveraged structured prompting for an LLM-based feature extraction pipeline to transform unstructured transaction metadata into high-signal categorical features, identifying subtle fraud markers that improved model precision by 8% in production. - Engineered multimodal Spark pipelines for high-volume telemetry, optimizing ensemble models (XGBoost, NNs) to achieve a 22% increase in PR-AUC and a 15% lift in Recall at a 0.1% FPR, while maintaining less than 10ms latency. - Designed and scaled a production MLOps framework on AWS for 90+ engineers, automating feature pipelines, distributed training, and CI/CD workflows to accelerate model rollout and ensure statistical reproducibility. - Led CECL production readiness assessments, performing controls and line-of-business requirement analyses to ensure compliance and operational stability.	
US Food and Drug Administration , Software Developer	Oct 2022 - June 2024
Developed an immersive regulatory science tool (Unity, WebXR, Javascript, ThreeJS) to evaluate contrast sensitivity in medical images on VR devices such as Meta Quest 2 and 3. (Publications below)	
Goldman Sachs , Data Science Intern	June - Aug 2023
Built scalable data and feature pipelines supporting ML experimentation across 150k+ trading portfolios, delivering real-time, structured datasets used for model evaluation and training analysis across 19 business lines.	
Goldman Sachs , Software Engineer Intern	June - Aug 2022
- Architected a 360-degree observability and audit layer for the Core Payments team, utilizing Terraform to automate the deployment of containerized (Podman) logging that ingest SLF4J telemetry into Elasticsearch for real-time transaction monitoring. - Collaboratively deployed a microservices architecture application (SpringBoot framework/Java, NoSQL queries from non-relational MongoDB) on Kubernetes, to decouple services, enhance reusability, and ensure high availability for app outage notifications.	

PROJECTS

AI Agents on ARC with ElevenLabs and Circle - Link	Nov 2025
Engineered a multi-agent ML system across Cloudflare Workers and Arc network, integrating ElevenLabs voice AI for speech-driven task execution, LLM pdf parser, deterministic state machines, and USDC-settling smart contracts for end-to-end trade automation.	
Computer Vision - Link	May 2024
- Implemented advanced image processing using algorithms such as SIFT and Photometric Stereo for geometry recovery, robust image comparison, panoramic stitching, to enhance 3D rendering and recognition system with Python. - Validated performance using root mean square error, F1 score, invariance measurements, and image alignment and overlap quality metrics to ensure high accuracy and quality.	
MiniTorch	Aug - Dec 2023
Built a PyTorch ML library by translating mathematical models into algorithms for tensor languages, auto-differentiation, model debugging, testing, and visualization. Explored GPU fundamentals, compression techniques, CUDA, and low-power inference.	
First Year Research & Innovation	June - Aug 2020
- Implemented and trained a DCGAN generative model in PyTorch/TensorFlow and expanded it with stabilized training techniques used in modern diffusion/LLM systems such as adaptive optimizers, gradient penalties, curriculum training. - Built a training + evaluation pipeline for generative modeling, adding experiment tracking, custom metrics, and automated failure-case visualizations to guide iterative model improvements.	

CERTIFICATION/PUBLICATIONS

Bhansali, K., Lago, M. A., Beams, R., Zhao, C., & Badano, A. (2024). Evaluation of Monocular and Binocular Contrast Sensitivity on Virtual Reality Head-Mounted Displays . Optics Express.	AWS Certified Cloud Practitioner
Zhao, C., Bhansali, K., Beams, R., Lago, M. A., & Badano, A. (2024). Integrating Eye Rotation and Contrast Sensitivity into Image Quality Evaluation of Virtual Reality Head-Mounted Displays . Optics Express, 32, 24968–24984.	Microsoft Azure Fundamentals Certification
Bhansali, K., Lago, M. A., Beams, R., Zhao, C. (2024). Evaluation of Monocular and Binocular Contrast Perception on Virtual Reality Head-Mounted Displays . IEEE VR. pp. 951-952, doi: 10.1109/VRW62533.2024.00272.	Grace Hopper Celebration 2020, 2022 Scholarship
Poster @ AAAI Undergraduate Consortium	2023 Loreal Brandstorm USA Top 10 Finalist
	2022, 2020, 2019 OMSE Academic Excellence Award
	2022, 2020, 2019 Dean's List